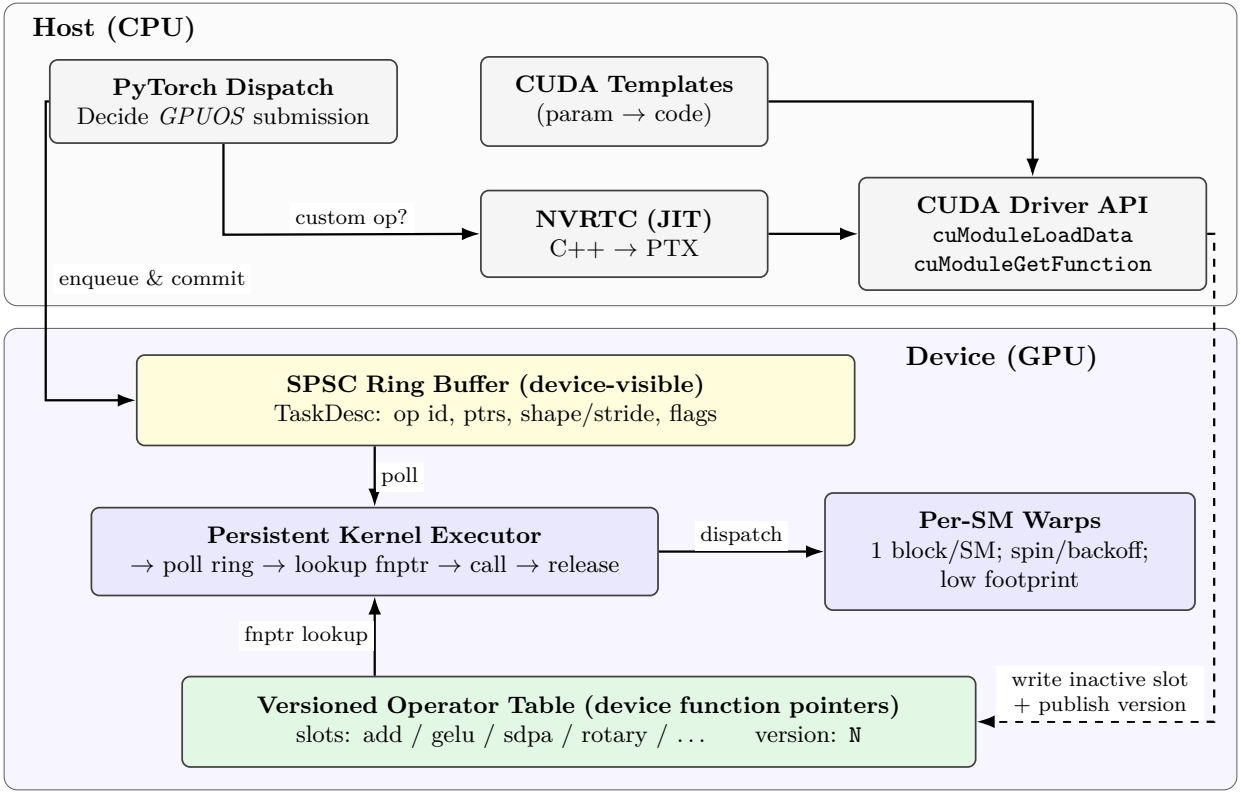

Host (CPU)
PyTorch Dispatch
Decide GPUOS submission
CUDA Templates
(param → code)
custom op?
NVRTC (JIT)
C++ → PTX
CUDA Driver API
cuModuleLoadData
cuModuleGetFunction
enqueue & commit
Device (GPU)
SPSC Ring Buffer (device-visible)
TaskDesc: op id, ptrs, shape/stride, flags
poll
Persistent Kernel Executor
→ poll ring → lookup fnptr → call → release
dispatch
Per-SM Warps
1 block/SM; spin/backoff;
low footprint
fnptr lookup
write inactive slot
+ publish version
Versioned Operator Table (device function pointers)
slots: add / gelu / sdpa / rotary / . . .
version: N